

AUTOMOTIVE SOFTWARE SHOULD-COST · COMPETITIVE BENCHMARK

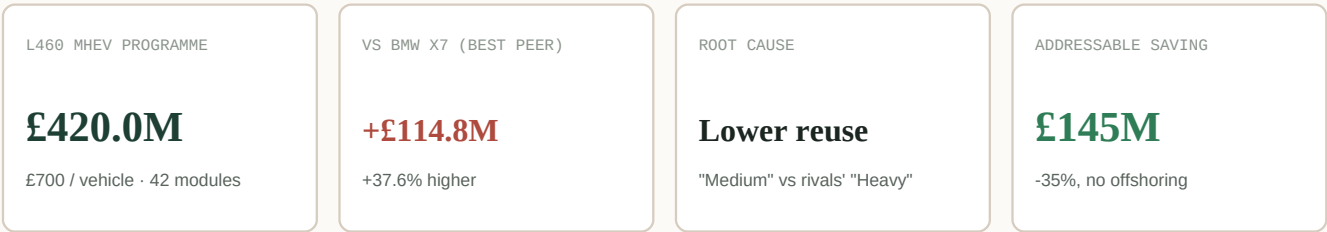
Range Rover L460 software cost benchmark

A like-for-like assessment of embedded-software programme cost against the premium-SUV peer set — with the optimisation opportunities available to close the gap.

BASELINE PROGRAMME £420M	GAP VS BEST PEER +38%	OPPORTUNITY IDENTIFIED £145M
--	---	--

01 The headline

On its core mild-hybrid (MHEV) programme — the only drivetrain the L460 shares with the BMW X7, Audi Q8 and Mercedes GLS — the L460's embedded software costs materially more than the German rivals. The cause is specific, and the fix is largely architectural rather than commercial.



Five findings

- 1

The L460's software costs ~38% more than the BMW X7 / Audi Q8 on the equivalent MHEV programme.

£420M vs £305M / £313M. Per vehicle the gap is smaller (£700 vs £636 / £631) — the L460 amortises over higher volume.
- 2

The gap is a software-reuse gap, not a geography or volume gap.

Normalised for region, source and volume, the L460 still sits ~£155M above the Germans — because BMW and Audi carry far more software forward between models.
- 3

The L460's commercial setup is already efficient.

Its Tier-1 supplier sourcing (0.88x) and UK base are lowering cost — so the opportunity is engineering strategy, not procurement.
- 4

Lifting reuse to peer level saves ~£145M (-35%) and reaches parity — no offshoring.

Raising baseline reuse "Medium to Heavy" plus shared-platform middleware brings the programme to ~£275M.
- 5

On electrified variants the L460 already benchmarks favourably.

Against the Porsche Cayenne (only peer with PHEV/BEV) the L460 is cheaper on both total and per-vehicle.

02 What can honestly be compared

A credible benchmark only compares drivetrains that both cars actually sell. The peer set does not offer every propulsion — so we compare only where a real, production, apple-to-apple pairing exists. Availability verified against manufacturer / industry sources (July 2026).

	ICE (non-hybrid)	MHEV (48V)	PHEV	BEV
Range Rover L460 reference — "our car"	—	●	●	🌀 launching
BMW X7 (G07) iX7 BEV is next-gen, 2027+	—	●	—	—
Audi Q8 Q8 e-tron is a separate model	—	●	—	—
Mercedes GLS EQS SUV is a separate model	—	●	—	—
Porsche Cayenne Electric premiered Nov 2025	●	—	●	●

●

 Offered in production

🌀

 Launching (Range Rover Electric)

—

 Not offered

Valid apple-to-apple pairings

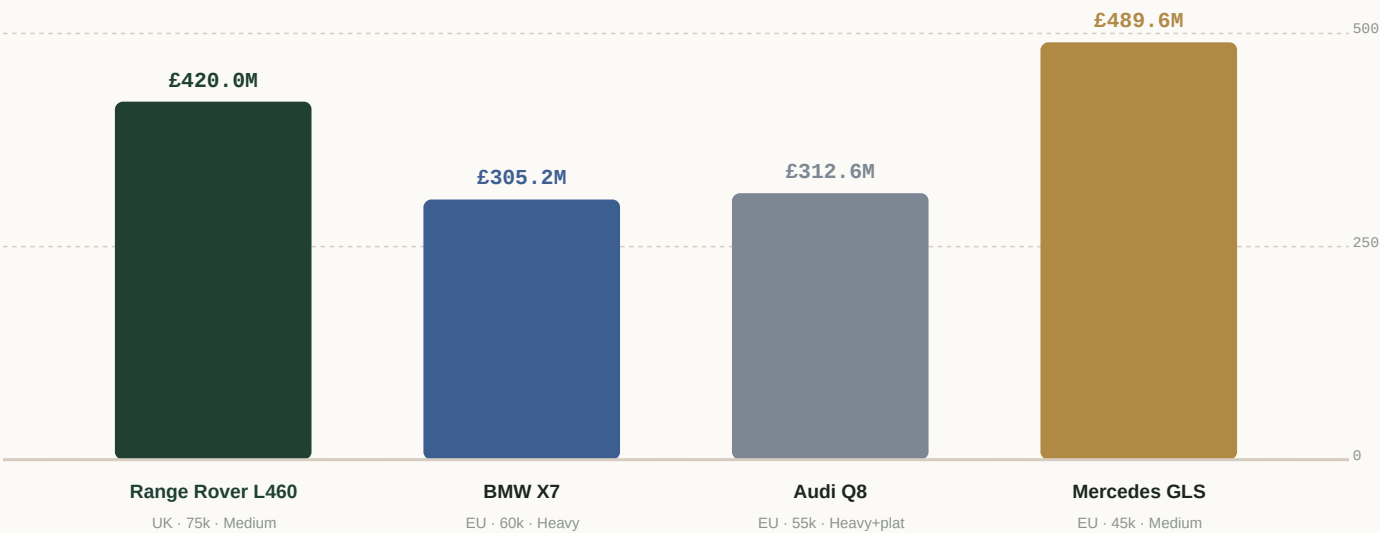
- MHEV: L460 vs X7, Q8, GLS — the four-way core benchmark.
- PHEV & BEV: L460 vs Porsche Cayenne — the only peer offering electrified variants.

Comparing the L460's software against, say, a "BMW X7 BEV" would be meaningless — BMW does not build one on this platform. Restricting to real pairings keeps every number defensible in front of a supplier or the board.

03 MHEV programme, four-way

Total embedded-software should-cost for the 48V mild-hybrid programme, each car on its own real commercial assumptions. The L460 is the second-most expensive of the four, and 38% above the leanest.

Total software programme cost — MHEV (£M) lower is better



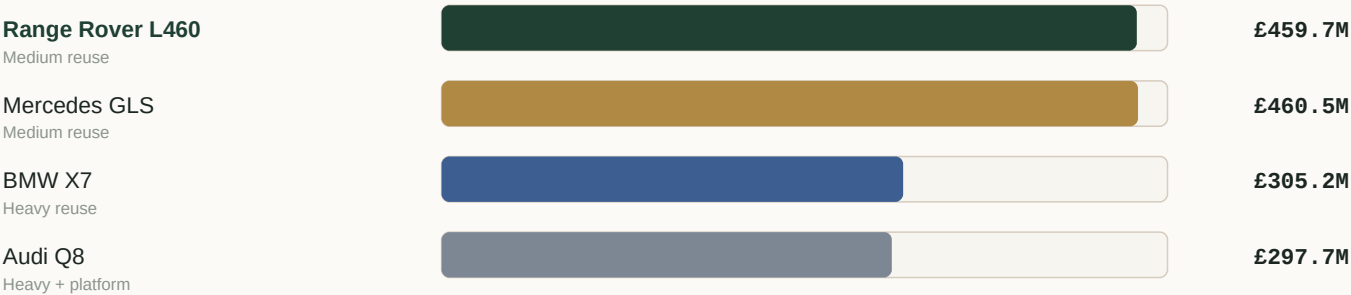
PROGRAMME (MHEV)	TOTAL	NRE	£/VEHICLE	vs L460
Range Rover L460	£420.0M	£253.6M	£700	—
BMW X7	£305.2M	£174.7M	£636	-£114.8M
Audi Q8	£312.6M	£168.9M	£631	-£107.4M
Mercedes GLS	£489.6M	£286.6M	£1,209	+£69.6M

Per-vehicle differences also reflect volume: the GLS is dearest per car (45k/yr) despite the L460's higher total. The L460's £700/vehicle is competitive — its total programme cost is the concern.

04 Where the difference comes from

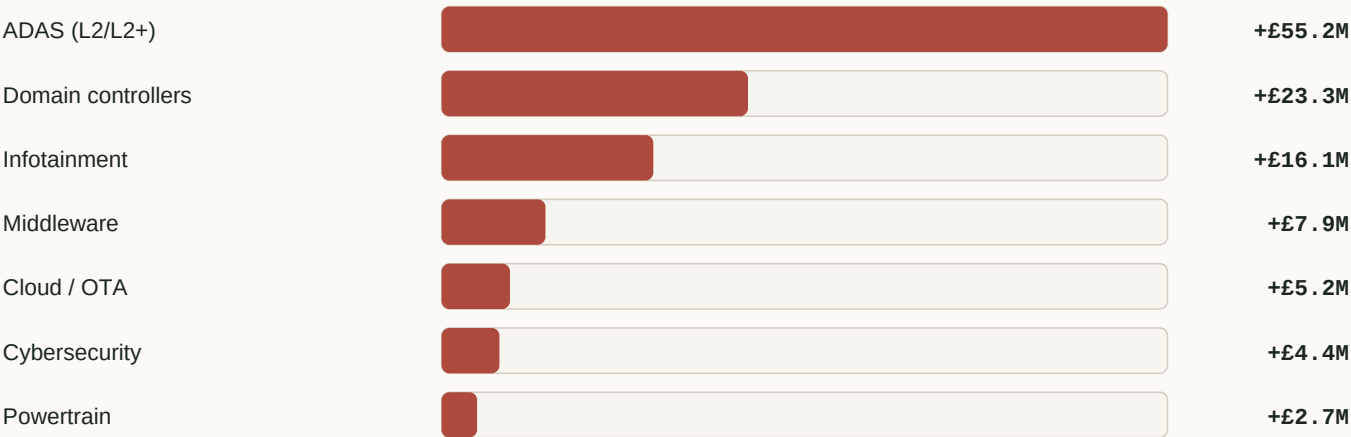
Two tests isolate the cause. A normalised view removes commercial/geographic differences; a domain breakdown shows which software carries the excess.

Normalised — same region, source & volume (£M)



Even on identical commercial terms the L460 stays high — clustering with the GLS (both "Medium") and well above BMW/Audi ("Heavy"). The gap is engineering strategy, not geography.

L460 excess over BMW X7, by domain (£M)



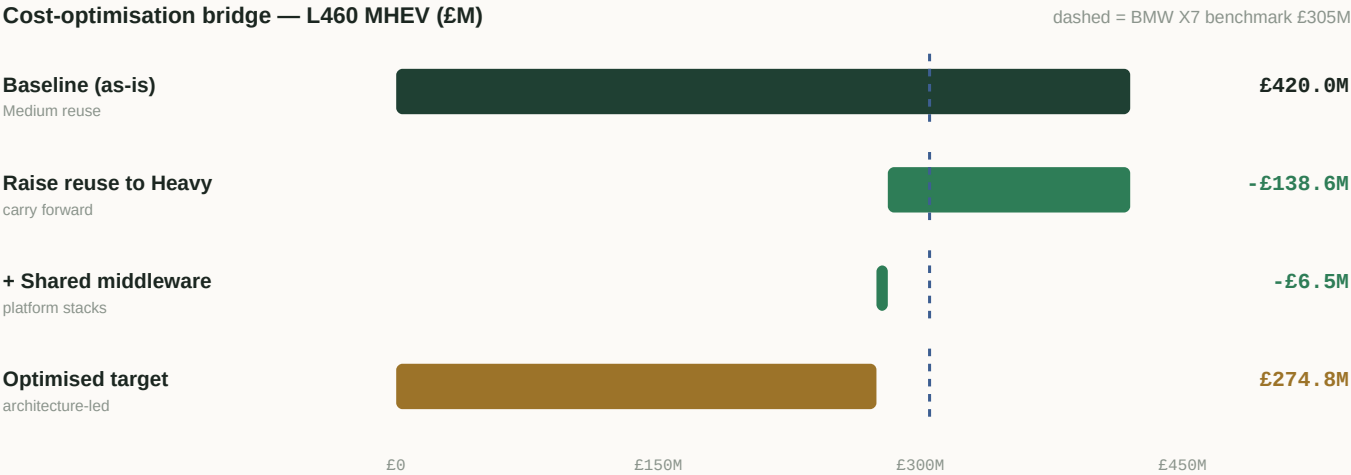
ADAS is half the excess (+£55M): it is the largest domain, so the lower reuse rate compounds there most.

The mechanism.

Reuse "Medium" applies 0.60× to development effort; "Heavy" applies 0.35×. BMW and Audi carry far more forward, re-developing ~half as much per programme. Audi additionally runs shared VW-Group middleware at "Platform" reuse (0.14×).

05 Cost-optimisation opportunities

The engine quantifies each lever on the L460 MHEV programme. The architecture-led path — more reuse + shared middleware — reaches German parity without changing where or by whom the work is done.



Bottom line

The two architectural levers take the L460 MHEV programme from £420M to £275M — a £145M (35%) saving, moving it from 38% above the X7 to roughly on par. Achievable in the UK, with the existing Tier-1 model.

The levers, ranked

- 1 · Raise baseline reuse to "Heavy"** -£138.6M

Match BMW's carry-forward across Range Rover / Sport / Velar.

RECOMMENDED · HIGHEST IMPACT
- 2 · Shared-platform middleware** -£6.5M*

Run AUTOSAR / RTOS / comms as a shared platform (as Audi does).

RECOMMENDED
- 3 · Nearshore a dev share** up to -£191M

Large nominal lever but IP / quality / continuity risk. Strategic, not first.

HANDLE WITH CARE
- 4 · ASIL right-sizing** review

Ensure no module exceeds the safety level its function requires.

QUICK WIN

06 Electrified & full ranking

The Cayenne is the only peer offering PHEV and BEV. Here the L460 is the cheaper programme on both total and per-vehicle — its software organisation is not structurally expensive.

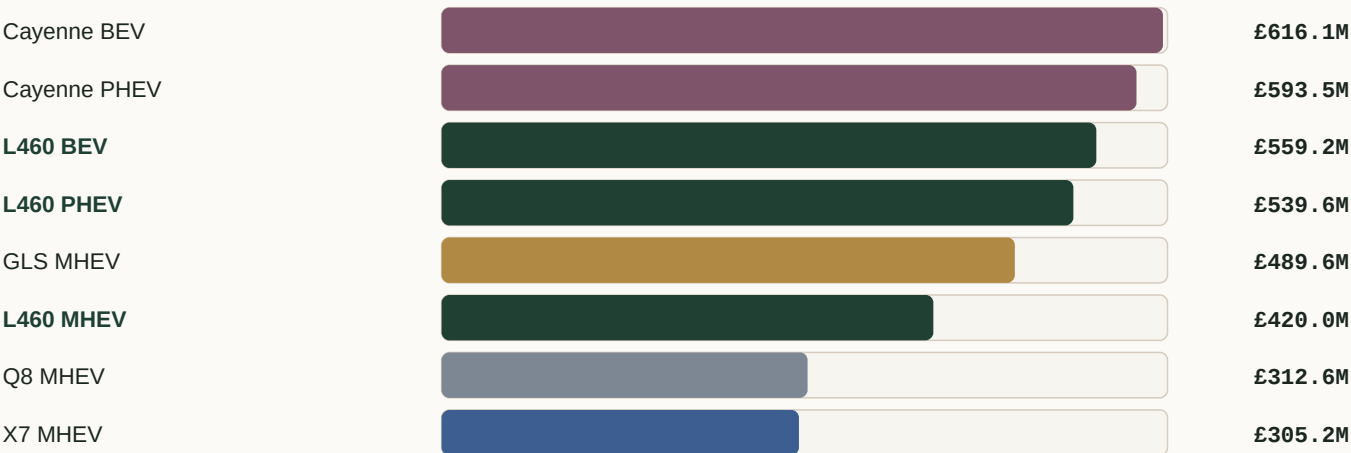
Electrified programme cost — L460 vs Porsche Cayenne (£M)



Read-across

On electrified programmes the L460 out-benchmarks a Porsche (~9% on both PHEV and BEV; £899 vs £1,484 and £932 vs £1,540 per vehicle). The MHEV cost problem is specific and reuse-driven — which is why it is fixable.

Full ranking — all comparable programmes, total software cost (£M)



After the £145M optimisation, the L460 MHEV would fall to ~£275M — the leanest programme in the set.

07 Assumptions & methodology

Every figure is live output from the CostVision should-cost engine (49-module catalogue, ISO 26262 / 21434). Nothing is hand-entered. Assumptions are stated in full so each number can be challenged.

Model class & accuracy

Bottom-up parametric should-cost; validated envelope $\pm 25\text{--}35\%$ (back-test MAPE 24.9%, 5 of 7 published programmes within $\pm 35\%$). Absolute numbers are estimates; the comparisons are consistent because every programme uses the same engine and rate library.

Propulsion availability

Verified July 2026. X7 (G07) MHEV-only (iX7 is next-gen 2027+); Q8 & GLS MHEV-only (e-tron / EQS are separate models); Cayenne offers ICE/PHEV/BEV (Electric premiered Nov 2025); Range Rover Electric is JLR's first EV, launching.

"Apple-to-apple"

Two views — as-is (each car on its real region, volume, life, reuse) and normalised (common EU / in-house / 60k-yr basis). Comparisons drawn only where both cars build the drivetrain.

Reuse factors ($\times$ dev effort)

Fresh 1.00 · Light 0.82 · Medium 0.60 · Heavy 0.35 · Platform 0.14. "Baseline reuse" is the programme default before per-module overrides.

Per-car assumptions (MHEV)

L460: UK · Tier-1 · 75k/yr · 8yr · Medium. X7: EU · in-house · 60k · 8yr · Heavy. Q8: EU · in-house · 55k · 9yr · Heavy + platform. GLS: EU · in-house · 45k · 9yr · Medium. Cayenne: EU · in-house · 50k · 8yr · Medium + platform + Porsche performance.

Signature software

Each car keeps its brand-DNA overrides (Meridian on L460; MBUX/voice on GLS; Porsche 4D chassis on Cayenne; UWB key on X7), held constant across its drivetrains.

What "saving" means

Reduction in total programme software should-cost (NRE + lifecycle) vs the L460 as-is baseline, from changing the stated lever only.

Currency & scope

All figures GBP. Embedded & connected vehicle software only; excludes manufacturing, hardware and non-software engineering.

This report supports target-setting and supplier negotiation; it is not an audited actual. Engine: CostVision sw-should-cost.ts, rate library v1.0, July 2026.